A reproducible global map of Neptune from Voyager 2 narrow-angle camera images

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Technical note on the Neptune texture of the Realistic Planets mod for OpenReliant

October 2026

Abstract

We describe the construction of a global map of the albedo and colour of Neptune from 34 calibrated images taken by the narrow-angle camera of Voyager 2 on 17 and 20 August 1989. The map was made as a texture for the Realistic Planets modification of OpenReliant, an open-source reimplementation of the video game StarLancer (1999), but is intended to be of use beyond it. As no reconstructed pointing of the cameras was available to us, the pointing of each image was determined from the image itself: the position and scale of the disk from a robust fit to the sunlit limb, with a median residual of 0.47 pixels, and the roll angle from the requirement that the cloud bands follow parallels of latitude, pooled over the images of each day, which gives $-6.85°$ and $-6.31°$ with bootstrap standard deviations of $0.47°$ and $0.25°$. The images were corrected for limb darkening with Minnaert's law and for fixed-pattern blemishes of the camera, advected to a common epoch with the zonal wind profile of Sromovsky et al. (1993), cross-calibrated photometrically through their overlaps, and mosaicked on a grid of $0.1°$. The colour was derived from the geometric albedo spectrum of Karkoschka (1994), modulated by mosaics made in the orange, green and blue filters, and converted to sRGB with the CIE 1931 colour-matching functions; the mean colour of the disk is sRGB (138, 182, 202). The map contains data over 85 % of the surface, from the south pole to about 40°N, at a resolution of 62 to 70 km per pixel; the remaining northern latitudes are filled with mirrored southern bands. The individual methods are well established, and Voyager mosaics of Neptune at higher resolution exist; the contribution of this note is a documented, reproducible processing pipeline and a map released under an open licence.

Keywords: Neptune; Voyager 2; image mosaicking; planetary photometry; true colour

1 Introduction

StarLancer (1999) represents Neptune by a sphere of 720 triangles, or 320 at large distances, with a texture of 256×256 pixels. The Realistic Planets modification for OpenReliant replaces it with a sphere of 16,128 triangles and a map derived from spacecraft images. The map has to remain sharp at large angular sizes: at one navigation point of the game the viewer is 0.59 million game units above the cloud tops, where the planet subtends about 86°.

Neptune has been mapped from Voyager 2 images since the encounter. The first results of the imaging team described the planet's bands and discrete features (Smith et al., 1989), and the motions of its clouds were measured in detail by Limaye & Sromovsky (1991) and Sromovsky et al. (1993). The mosaics produced by B. Jónsson in 2014 from about 20 green- and clear-filter images taken at ranges of 3 to 4 million km are among the most detailed global views of the planet made since (Lakdawalla, 2014); no licence for their reuse is stated. The approach described here follows the same general lines: calibrated, geometrically corrected images are projected onto a latitude-longitude grid, with the illumination removed by a photometric model. Irwin et al. (2024) used spectra from the Hubble Space Telescope and the Very Large Telescope to rebalance colour

images from Voyager 2 and Hubble, and showed that the true colour of Neptune is a pale greenish blue, only slightly bluer than that of Uranus. Hubble's Outer Planet Atmospheres Legacy (OPAL) programme maps Neptune regularly at two pixels per degree (Simon et al., 2015; OPAL Team, 2015).

For use in the game, the map was required to be global, sharp at large angular sizes, close to true colour, and available under an open licence. None of the products known to us met all four requirements: the Hubble maps resolve about 1000 km on Neptune, no licence is stated for Jónsson's mosaics, and the openly licensed maps of Neptune that we found are partly painted. We therefore made a new map from the calibrated Voyager 2 images, which are in the public domain.

Section 2 describes the data, Section 3 the methods, Section 4 the results, and Section 5 the adjustments made for the game. Section 6 discusses the limitations of the map, and Section 7 concludes.

2 Data

2.1 Voyager 2 images

The images were obtained from the PDS Ring-Moon Systems Node through its search service, OPUS, which also provides the range, sub-spacecraft point and sub-solar point of each image. OPUS lists 716 narrow-angle camera images of Neptune taken between 14 and 25 August 1989. We used the GEOMED products of the data set VG2-N-ISS-2/3/4/6-PROCESSED-V1.0, which are radiometrically calibrated to I/F (the calibration of the cameras is described by Danielson et al., 1981) and geometrically corrected by resampling onto 1000×1000 pixels of 7.842 µrad.

Two sets of images were selected (Table 1). The luminance set consists of 10 clear- and green-filter images taken on 20 August, which cover all longitudes at 62 to 70 km per pixel at the centre of the disk. The colour set consists of eight triplets of orange-, green- and blue-filter images taken on 17 August at 97 to 109 km per pixel. In all 34 images the phase angle is 15.0°, the planetocentric latitude of the sub-spacecraft point is −28.5° and that of the sub-solar point is −23.2°, so that the southern hemisphere is well observed, while high northern latitudes are seen obliquely or not at all. The whole disk fits within the frame at ranges beyond about 6.3 million km; the images of 20 August are among the last in which it does, which matters for the determination of the pointing (Section 3.2).

Table 1: The images used. Times are UTC; the resolution is that of a raw pixel at the centre of the disk. The individual images and their geometry are listed in `frames.json` in the sources of the map.

Set	Date (1989) and time	km per pixel	Images
Luminance	20 August, 00:31 to 15:13	62 to 70	7 clear, 3 green
Colour	17 August, 02:35 to 23:15	97 to 109	8 orange, 8 green, 8 blue

2.2 Reference spectrum

The colour of the map is based on the geometric albedo spectrum of Neptune measured by Karkoschka (1994) at the European Southern Observatory from 23 to 26 July 1993, which covers 300 to 1000 nm sampled at intervals of 0.4 nm. It is archived in the data set ESO-J/S/N/U-SPECTROPHOTOMETER-4-V2.0 at the PDS Atmospheres Node (Karkoschka, 1998).

2.3 Zonal winds

The clouds of Neptune move rapidly relative to the planet's rotation (System III), at about $400\,\mathrm{m\,s^{-1}}$ westward at the equator. We used the symmetric polynomial fitted by Sromovsky

et al. (1993) to the cloud motions measured in Voyager images,

$$U(\varphi) = -398 + 0.188\,\varphi^2 - 1.2 \times 10^{-5}\,\varphi^4 \quad \mathrm{m\,s^{-1}}, \tag{1}$$

where φ is the planetographic latitude in degrees and U is positive eastward.

3 Methods

3.1 Geometric model

Neptune is modelled as an oblate spheroid,

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{c^2} = 1, \qquad a = 24{,}764\,\mathrm{km}, \quad c = 24{,}341\,\mathrm{km}, \tag{2}$$

in body-fixed coordinates with z along the spin axis (Archinal et al., 2018). Let $\mathbf{O}$ be the position of the spacecraft, $\hat{\mathbf{f}}$ the unit vector from the spacecraft towards the centre of Neptune, $\hat{\mathbf{n}}$ the unit vector along the projection of the spin axis onto the plane normal to $\hat{\mathbf{f}}$, and $\hat{\mathbf{r}} = \hat{\mathbf{f}} \times \hat{\mathbf{n}}$. A point $\mathbf{P}$ on the planet, with $\mathbf{d} = \mathbf{P} - \mathbf{O}$, is imaged at sample u and line v (increasing downwards):

$$\xi = \frac{\mathbf{d} \cdot \hat{\mathbf{r}}}{\mathbf{d} \cdot \hat{\mathbf{f}}}, \qquad \eta = \frac{\mathbf{d} \cdot \hat{\mathbf{n}}}{\mathbf{d} \cdot \hat{\mathbf{f}}}, \tag{3}$$

$$u = u_0 + \frac{s}{p}\left(\xi \cos\theta - \eta \sin\theta\right), \qquad v = v_0 - \frac{s}{p}\left(\xi \sin\theta + \eta \cos\theta\right). \tag{4}$$

Here $p = 7.842\,\mu\mathrm{rad}$ is the nominal pixel scale of the GEOMED products, and four parameters are fitted to each image: the position (u_0, v_0) of the centre of the planet, a scale factor s close to unity, and the roll angle θ, by which the projected spin axis is rotated counter-clockwise from the upward direction of the image. The inverse mapping intersects the ray through each pixel with the spheroid. The cosines of the emission and incidence angles, μ and μ_0, are computed from the normal of the spheroid.

3.2 Disk centre and scale

Reconstructed pointing of the Voyager 2 scan platform during the Neptune encounter was not available to us: the C-kernel distributed by NASA's Navigation and Ancillary Information Facility (NAIF) describes the attitude of the spacecraft bus, not of the scan platform that carries the cameras. The pointing was therefore determined from each image. An initial estimate of (u_0, v_0, s) was obtained by fitting a model disk with Minnaert limb darkening, including the terminator, to the image for a coarse set of roll angles. This estimate proved to be biased by up to 26 pixels, because the model is least accurate near the limb.

The final estimate uses the sunlit limb alone. Along radial lines at intervals of 1°, where the Sun is at least 8.6° above the horizon ($\mu_0 > 0.15$) at 97 % of the model limb radius, the brightness profile is sampled every 0.25 pixels; the limb point $\mathbf{q}_i$ is located, by linear interpolation, where the profile falls to the mean of the median brightness just inside the limb (0.90 to 0.96 limb radii) and that of the sky (beyond 1.06 limb radii). The parameters are then found by minimising

$$\sum_i \rho(\|\mathbf{q}_i - \mathbf{c}\| - R(\phi_i;\, \mathbf{c}, s, \theta)), \qquad \rho(z) = 2\left(\sqrt{1 + z^2} - 1\right), \tag{5}$$

where $\mathbf{c} = (u_0, v_0)$, ϕ_i is the direction of $\mathbf{q}_i$ from $\mathbf{c}$, R is the distance of the model limb from $\mathbf{c}$ in that direction, in pixels, and ρ is a robust loss with a scale of one pixel. The limb points are detected again with the updated model, three times in all.

3.3 Roll angle

The outline of the disk constrains the roll angle only weakly: at a phase angle of 15° it is nearly circular, and the quality of the limb fit changes by less than 1 % over a range of 30° in θ. The roll angle was instead determined from the cloud bands, which on Neptune follow parallels of latitude. For a trial angle θ, the image is projected onto a grid of 0.5° in latitude and longitude between $-72°$ and 30° latitude, using only cells with $\mu, \mu_0 > 0.45$, and the albedo (Equation 8) is divided by its median $\tilde{A}$ over the image. The cost

$$C(\theta) = \frac{\sum_j n_j \, M_j(\theta)^2}{\sum_j n_j}, \qquad M_j = \operatorname*{median}_k \left| \frac{A_{jk}}{\tilde{A}} - \operatorname*{median}_k \frac{A_{jk}}{\tilde{A}} \right|, \tag{6}$$

is the mean squared median absolute deviation of the albedo along the latitude rows j, each weighted by its number n_j of valid cells k; rows with fewer than 60 valid cells are excluded. The median absolute deviation was preferred to the variance, which is dominated by discrete features such as the Great Dark Spot.

The minima of single images scatter by several degrees. Since the roll of the camera relative to Neptune is expected to change little within a day, the curves were evaluated from $-25°$ to 10° in steps of 0.5°, divided by their means, and summed over the images of each day. The roll angle of the day is the vertex of the parabola through the lowest sample of the summed curve and its two neighbours. Its statistical uncertainty was estimated by a bootstrap over the images of each day, with 2000 resamplings. The limb fit (Equation 5) was then repeated with the roll angle of the day.

3.4 Fixed-pattern blemishes

The first mosaics showed small dark rings that remain at the same pixel positions in every image, whereas the clouds do not. To isolate them, the band-pass-filtered brightness of each image (a Gaussian of 1.5 pixels minus one of 10 pixels, divided by the latter) was averaged in camera coordinates over the disks of all 34 images (Figure 3). Five compact rings were identified with a Laplacian-of-Gaussian filter and confirmed visually. Within 26 pixels of the centre of each, every image is divided by one plus this averaged pattern. Masking the rings instead left gaps in the mosaic where only one image observed a region well. Isolated deviant pixels, differing from the median of their 3×3 neighbourhood by more than 0.02 in I/F, are replaced by that median.

3.5 Photometric model

Limb darkening is described by Minnaert's law (Minnaert, 1941),

$$\frac{I}{F} = B \, \mu_0^k \, \mu^{k-1}, \tag{7}$$

fitted to each image by linear least squares in the form $\ln(\mu \, I/F) = \ln B + k \ln(\mu\mu_0)$, using pixels with $\mu, \mu_0 > 0.25$. The albedo of a point is then

$$A = \frac{I/F}{\mu_0^k \, \mu^{k-1}}. \tag{8}$$

For the luminance set, A is divided by the fitted B of each image, so that clear- and green-filter images can be combined; for the colour sets, B is retained, since the ratios between the filters carry the colour.

3.6 Photometric cross-calibration

Equation (7) is least accurate at large emission and incidence angles, and north of the equator all images observe the clouds obliquely. The first mosaics therefore showed steps in brightness

where one image took over from another. Where images overlap, the same location is observed at different angles, which allows a cross-calibration. For each pair of images (i, j) with at least 2000 common grid cells where $\mu, \mu_0 > 0.3$ in both, up to 20,000 cells were drawn at random, and the model

$$\ln \frac{A_i}{A_j} = g_i - g_j + [\mathbf{b}(\mu_i, \mu_{0,i}) - \mathbf{b}(\mu_j, \mu_{0,j})] \cdot \boldsymbol{\beta}, \qquad \mathbf{b}(\mu, \mu_0) = \left(\bar{\mu},\ \bar{\mu}^2,\ \bar{\mu}_0,\ \bar{\mu}_0^2,\ \bar{\mu}\,\bar{\mu}_0\right), \tag{9}$$

with $\bar{\mu} = 1 - \mu$ and $\bar{\mu}_0 = 1 - \mu_0$, was fitted by least squares, subject to $\sum_i g_i = 0$. Here g_i is a gain for each image and $\boldsymbol{\beta}$ the coefficients of an empirical correction to Equation (7) shared by all images of a set. Samples with residuals larger than four times the robust standard deviation (1.4826 times the median absolute deviation) were excluded and the fit repeated, four times in all, so that clouds that moved differently from Equation (1) do not bias the solution. Each image is then divided by $\exp(g_i + \mathbf{b} \cdot \boldsymbol{\beta})$.

3.7 Temporal registration and mosaicking

Each image is projected onto a grid of 0.1° (luminance) or 0.25° (colour) in planetographic latitude and east longitude, after the clouds at each latitude have been moved to the epoch of 20 August 1989, 08:00 UTC, by

$$\Delta\lambda = \frac{U(\varphi)\,\Delta t}{N(\varphi)\cos\varphi}, \qquad N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2\varphi}}, \qquad e^2 = 1 - \frac{c^2}{a^2}, \tag{10}$$

where Δt is the time from the image to the epoch and $N\cos\varphi$ the distance from the spin axis. Cells with μ or μ_0 below 0.25, or within 6 pixels of the edge of the valid image area, are discarded. Each remaining cell of image i receives the weight

$$w_i = \left[\frac{(\mu\,\mu_0)^2\, e_i}{r_i^2}\right]^3, \qquad e_i = \min\left(1, \frac{D_i}{40}\right), \tag{11}$$

where r_i is the resolution of the image in km per pixel and D_i the distance of the cell from the edge of the valid image area, in pixels. The mosaic is the weighted mean of the calibrated albedos,

$$\bar{A} = \frac{\sum_i w_i\, A_i\, \exp(-g_i - \mathbf{b}_i \cdot \boldsymbol{\beta})}{\sum_i w_i}. \tag{12}$$

The cube in Equation (11) lets the best observation of each location dominate without hard seams, and limits the averaging of observations made at different times, which would otherwise double features whose motion departs from Equation (1).

3.8 Colour

The orange, green and blue filters of the narrow-angle camera have effective wavelengths of about 591, 566 and 479 nm. They do not sample the red part of the spectrum, where absorption by methane is strongest and which strongly affects the colour of Neptune, so they cannot provide its true colour on their own. The mean colour was therefore taken from the reference spectrum $A(\lambda)$, and the filter mosaics were used only to modulate it. The linear sRGB colour of a reflectance spectrum is

$$\mathbf{c} = \sum_\lambda A(\lambda)\,\mathbf{m}(\lambda), \qquad \mathbf{m}(\lambda) = \mathsf{T}\,\frac{S(\lambda)\,\bar{\mathbf{x}}(\lambda)}{\sum_{\lambda'} S(\lambda')\,\bar{y}(\lambda')}, \tag{13}$$

where S is the spectrum of a black body at 5778 K standing in for sunlight, $\bar{\mathbf{x}} = (\bar{x}, \bar{y}, \bar{z})$ are the CIE 1931 colour-matching functions in the analytic form of Wyman et al. (2013), and T is the product of the matrix from CIE XYZ to linear sRGB (IEC, 1999) and a Bradford chromatic

adaptation (Lam, 1985) from the white point of S to D65, so that a perfect white reflector gives $(1, 1, 1)$. At each location $\mathbf{x}$ of the map, the colour is

$$\mathbf{c}(\mathbf{x}) = \sum_f q_f(\mathbf{x}) \sum_\lambda A(\lambda)\, h_f(\lambda)\, \mathbf{m}(\lambda), \tag{14}$$

where f runs over the blue, green and orange filters, q_f is the mosaic of filter f divided by its mean between 60°S and 20°N and smoothed with a Gaussian of standard deviation 2°, and the h_f are piecewise-linear functions that sum to one: h_B falls from 1 at 479 nm to 0 at 566 nm, h_O rises from 0 at 566 nm to 1 at 591 nm, and $h_\mathrm{G} = 1 - h_\mathrm{B} - h_\mathrm{O}$. The colour images were taken three days before the epoch, and the motions of discrete features depart from Equation (1) over that interval, which is why the colour is smoothed and the brightness is taken from the luminance mosaic alone. The texture is

$$\mathbf{t}(\mathbf{x}) = \frac{\mathbf{c}(\mathbf{x})}{Y(\mathbf{c}(\mathbf{x}))}\, Y(\bar{\mathbf{c}})\, \kappa\, \max\bigl(0.05,\ 1 + E\left[L'(\mathbf{x}) - 1\right]\bigr), \tag{15}$$

where $Y(\mathbf{c}) = 0.2126\, c_\mathrm{R} + 0.7152\, c_\mathrm{G} + 0.0722\, c_\mathrm{B}$ is the luminance, $\bar{\mathbf{c}}$ the colour of the unmodulated spectrum, L' the luminance mosaic divided by its mean between 60°S and 20°N and completed at high northern latitudes (Section 3.9), and E a contrast factor, equal to one for the true-colour map. The factor κ converts the geometric albedo of the spectrum to the Minnaert albedo of the mosaics: it is the ratio of the mean of the green mosaic between 60°S and 20°N to the spectrum's albedo in a Gaussian passband of standard deviation 17 nm centred on 566 nm. The result is resampled to 4096×2048 pixels with a Lanczos filter in linear light and encoded in 8-bit sRGB.

3.9 High northern latitudes

North of 32°N all images observe the clouds close to the limb, and the data end between 37°N and 47°N. Between 32°N and 40°N the relative luminance map L is blended into a mirror image of the southern hemisphere,

$$\begin{aligned} L'(\varphi, \lambda) &= [1 - \sigma(\varphi)]\, L(\varphi, \lambda) + \sigma(\varphi)\, \langle L(-\varphi, \cdot)\rangle_{45^\circ}, \\ \sigma &= t^2(3 - 2t), \qquad t = \min\!\left(1, \max\!\left(0, \frac{\varphi - 32^\circ}{8^\circ}\right)\right), \end{aligned} \tag{16}$$

where $\langle\cdot\rangle_{45^\circ}$ denotes a running mean over 45° of longitude, applied after a Gaussian smoothing of standard deviation 0.3°, so that no individual feature appears twice. Where L is undefined, $L' = \langle L(-\varphi, \cdot)\rangle_{45^\circ}$. The colour is blended likewise with its mirror image. This fill is a plausible extrapolation, not data.

4 Results

4.1 Pointing

The limb fits used 67 to 221 points per image, with root-mean-square residuals between 0.37 and 0.81 pixels and a median of 0.47 pixels (Figure 1). The scale factor is $s = 0.9976 \pm 0.0009$ (mean and standard deviation over the 34 images), with a range of 0.9949 to 0.9996. This small, consistent departure from unity absorbs any error in the nominal pixel scale as well as any systematic offset between the half-intensity limb and the model limb.

The summed roll-angle curves have well-defined minima (Figure 2). The roll angle is −6.85° on 17 August (24 images) and −6.31° on 20 August (10 images), with bootstrap standard deviations of 0.47° and 0.25°. For 23 of the 34 images, the minimum of the individual curve lies within 1° of the value of its day. The result is also sensitive to details of the processing: computing the Minnaert exponents with the biased initial disk centres instead of the limb-fitted ones changed the roll angles by 0.94° and 0.19°. We therefore regard the roll angles as uncertain by about 1°.

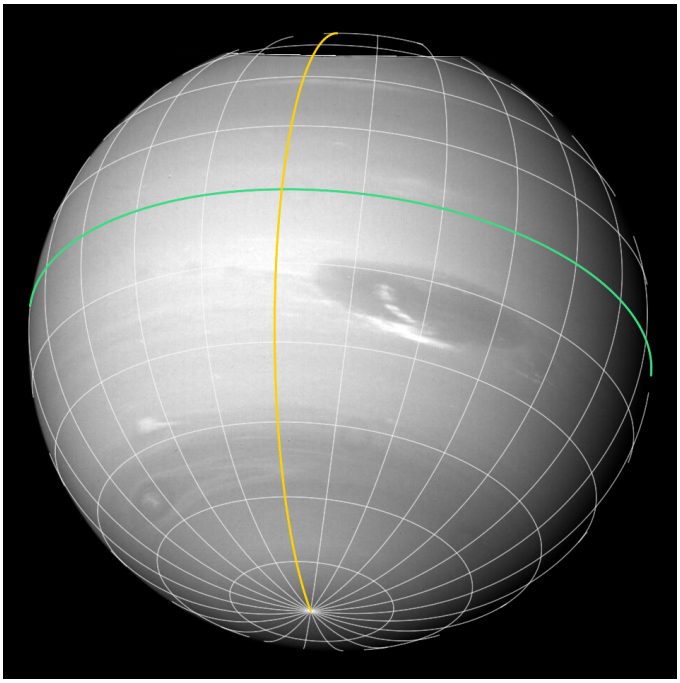


Figure 1: Clear-filter image C1124452, taken on 20 August 1989 at 66 km per pixel, with lines of latitude and longitude every 15° from the fitted pointing; the equator is shown in green and the meridian 0° in yellow. The disk centre and scale were fitted to the limb, the roll angle to the cloud bands. The Great Dark Spot is to the right of the centre. The brightness is displayed with a power of 1.8.

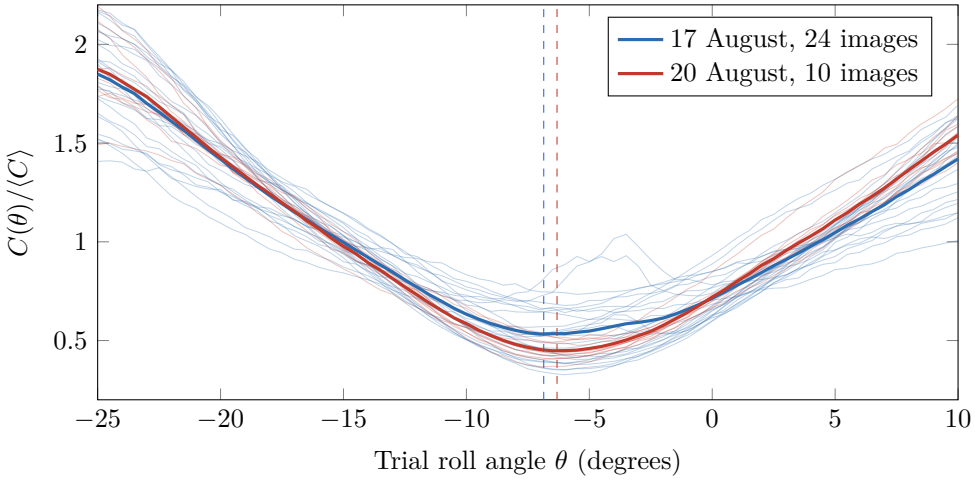


Figure 2: The roll-angle cost (Equation 6) of each image divided by its mean (thin lines), the mean over the images of each day (thick lines), and the roll angle adopted for each day (dashed lines).

4.2 Photometry

The fitted Minnaert parameters (Table 2) are consistent between images of the same filter, with standard deviations of 0.003 to 0.005 in k, and the exponent increases towards shorter wavelengths. The cross-calibration of the luminance set used 900,000 pairs of samples, of which 0.4 % were rejected, and the robust standard deviation of the residuals is 2.2 %. The fitted gains lie between 0.995 and 1.007, while the shared correction ranges from 0.942 to 1.035 between the 1st and 99th percentiles of the cells used; it is this angular correction that removes the steps in brightness between images (Figure 4).

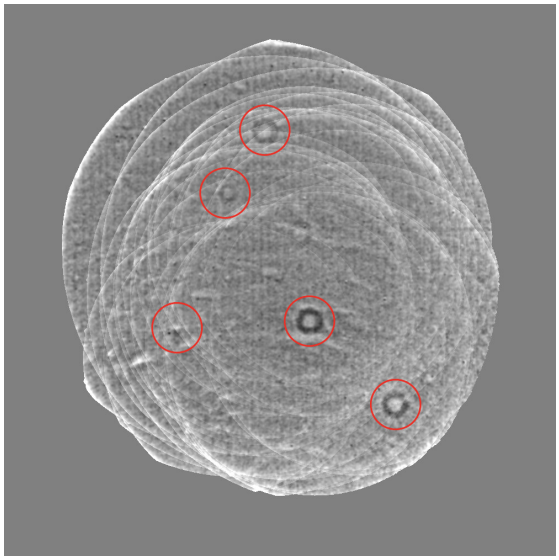


Figure 3: The band-pass-filtered brightness averaged in camera coordinates over all 34 images. The five fixed rings that are corrected are circled; the arcs are the limbs of the individual images.

Table 2: Minnaert parameters by filter: the mean exponent k with its standard deviation over the images, and the mean albedo B.

Filter	Images	k	B
Orange (591 nm)	8	0.680 ± 0.003	0.397
Green (566 nm)	11	0.712 ± 0.003	0.505
Clear	7	0.752 ± 0.003	0.592
Blue (479 nm)	8	0.768 ± 0.005	0.628

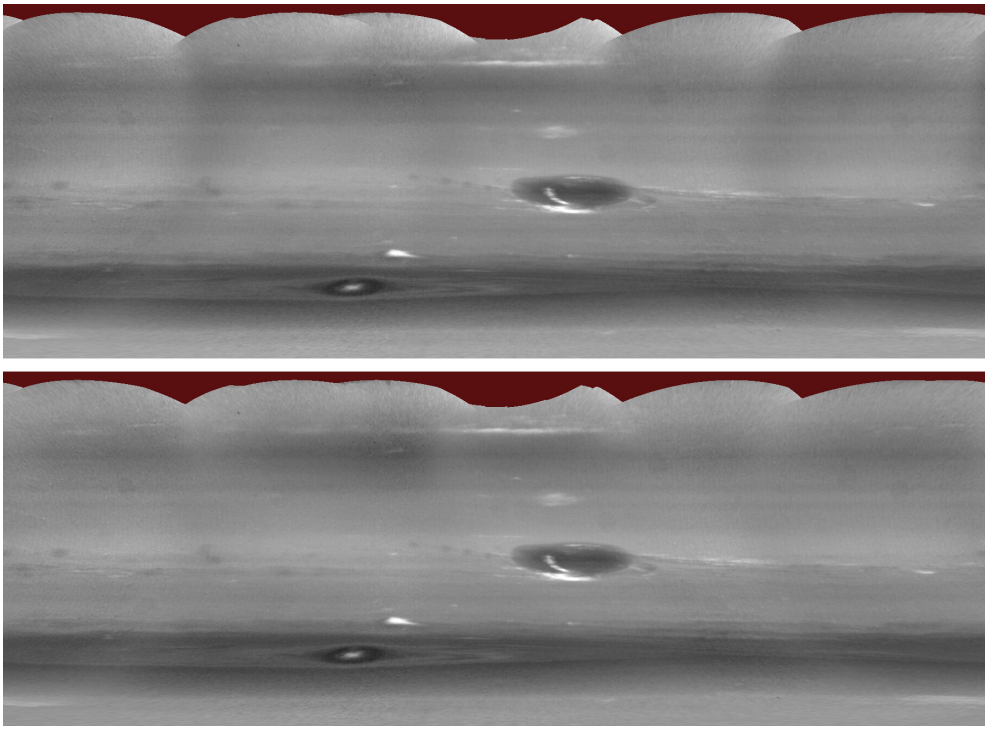


Figure 4: The luminance mosaic from 80°S to 50°N, with its contrast stretched to show errors, before (top) and after (bottom) the photometric cross-calibration. North is up and the meridian 0° is in the middle. Dark red marks the absence of data.

4.3 Colour

The unmodulated spectrum gives the linear sRGB colour $\bar{\mathbf{c}} = (0.252, 0.471, 0.589)$, or (138, 182, 202) in 8-bit sRGB: a pale, slightly greyish cyan blue, broadly consistent with the colour found by Irwin et al. (2024). As a check of consistency, the ratios between the filters were compared with those of the spectrum in Gaussian passbands of standard deviation 17 nm. The ratio of blue to green is 1.267 in the spectrum and 1.242 in the mean Minnaert albedos of the colour images, which agree to within 2 %; the ratio of orange to green is 0.862 and 0.786. The larger difference in orange may arise from the long-wavelength tail of the orange filter, which extends into the methane band at 619 nm, from the simplified passbands, and from changes in the planet between 1989 and 1993. Since the map takes only the variations of each filter about its own mean from the Voyager images (Equation 14), this difference does not affect its colour. The factor κ in Equation (15) is 1.155.

4.4 The map

The map contains data over 84.8 % of the surface. It is complete in longitude up to 36.8°N and has partial coverage up to 46.8°N. In true colour, the brightness varies little: between 70°S and 30°N the standard deviation of L is 3.8 %, and 96 % of the area lies between 8.2 % below and 5.9 % above the mean. The bands and the large discrete features observed by Voyager 2, including the Great Dark Spot and its bright companion clouds, the bright feature known as the Scooter, and the second dark spot with its bright core, are visible in the map (Figures 5 and 6).

5 Adaptation for StarLancer

Three adjustments were made for the game alone; they are not part of the measured map. First, a brightness variation of a few per cent appears featureless on a game screen, so the texture uses $E = 3$ in Equation (15), in the same spirit as the contrast stretches applied to published Voyager images of Neptune. The mean colour is unchanged by this. Second, the game illuminates a planet with a single light of colour $(1, 1, 0.8)$, which its tone curve turns into a pronounced yellow cast; the blue channel of the texture is therefore divided by 0.8. The texture is equirectangular, with north at the top, the meridian 0° in the middle column, and east to the right, as the spheres of the mod are mapped.

Third, the limb is rendered with the part of Minnaert's law that depends on the viewing geometry. A surface shader multiplies the texture, in linear light, by μ^{k-1} in each colour channel, with the exponents of Table 2: that of the orange filter for the red channel, for lack of a red filter, and those of the green and blue filters. Since the law was fitted only for $\mu \geq 0.25$, the factor is held constant closer to the limb, where it reaches about 1.5 in red and 1.4 in blue; the centre of the disk is unchanged. As k is smallest in red, the sunlit limb becomes both brighter and paler. The game's own illumination of the planet, the cosine of the incidence angle extended past the terminator, $\max(0, (\mu_0 + 0.25)/1.25)$, stands in for μ_0^k. It agrees with $\mu_0^{0.75}$ to within 1 % for $\mu_0 \geq 0.5$, but exceeds it by 13 % at $\mu_0 = 0.25$ and illuminates a band of about 14° beyond the terminator. Applying the full law would require the direction of illumination within the shader, which the engine does not yet provide.

6 Discussion

At 62 to 70 km per pixel, the map is coarser than the mosaics by Jónsson (Lakdawalla, 2014), which were made from images taken at 3 to 4 million km, but it is available under an open licence, and every step of its construction can be repeated from the archived data. Its main limitations are the following.

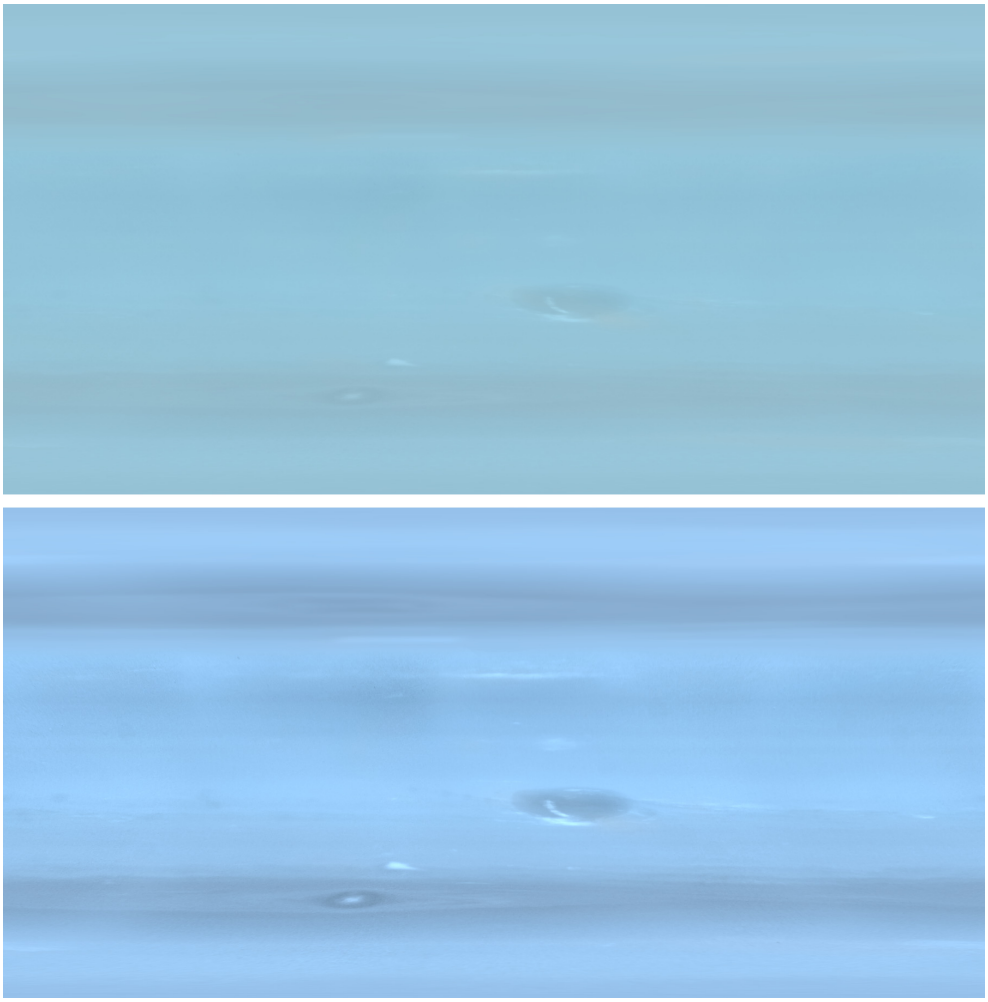


Figure 5: The map in true colour ($E = 1$, top) and as distributed for the game ($E = 3$, with the blue channel raised as described in Section 5, bottom). Equirectangular projection, north up, meridian 0° in the middle, east to the right.

High northern latitudes. North of 40°N the map contains mirrored southern bands (Equation 16) rather than data, and between 32°N and 40°N a blend of the two.

Different epochs. The luminance images span 15 hours and the colour images were taken three days earlier. Equation (1) describes the zonal mean motion; discrete features drift at their own rates and change shape within hours, so their positions, and the colour around them, are approximate. This applies to any mosaic of Neptune assembled from images taken over hours or days.

Images at higher resolution. Images from 21 and 22 August, at 30 to 53 km per pixel, could double the resolution. We attempted to include them, but most do not contain enough of the limb to fix their pointing, the roll angle appears to change by several degrees after 20 August, and the clouds had changed shape within a day or two; the resulting mosaic showed curved seams and mismatched features, and these images were left out. Including them would require the pointing of the scan platform, or mosaics of images taken close together in time, which would then be combined.

Roll angle. A single roll angle is assumed for each day, uncertain by about 1° (Section 4). An independent check against reconstructed pointing, should it become available, would be valuable.

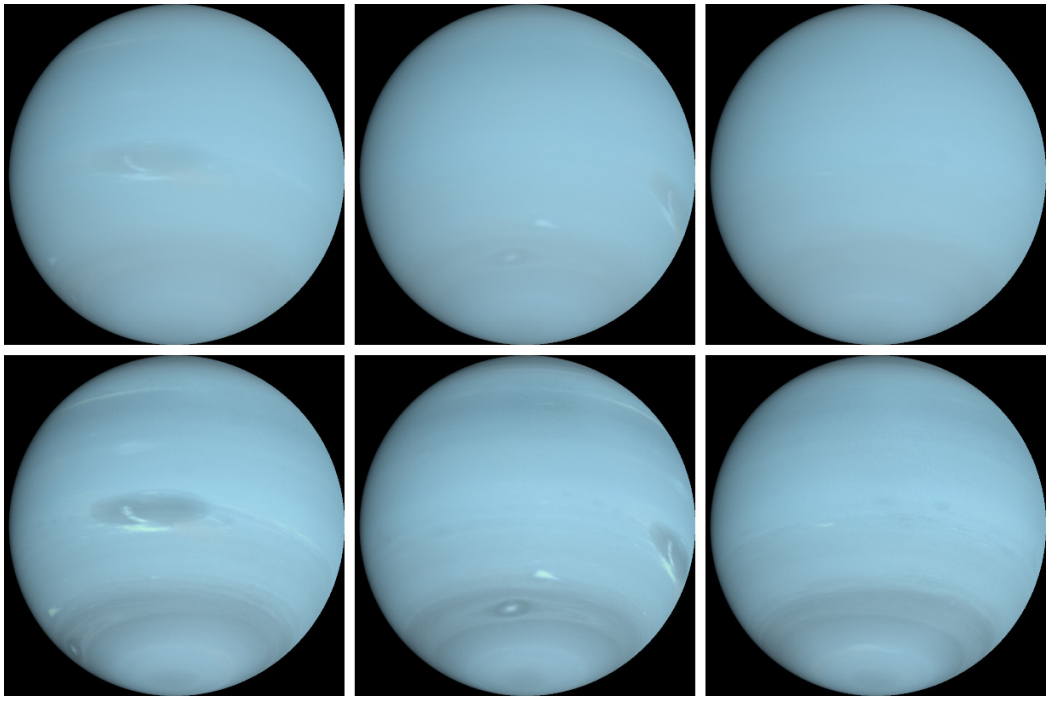


Figure 6: Orthographic views of the map with Minnaert shading ($k = 0.75$), seen from 25°S and illuminated from 15° east of the viewer, rendered by the script `render.py` of the sources and not by the game. Top: true colour under white light. Bottom: the texture as distributed, under the colour of the game's light, without the game's tone curve. Left: the Great Dark Spot and its companion clouds. Middle: the second dark spot and, above it, the Scooter. Right: the hemisphere away from both.

Photometry. Minnaert's law with a shared empirical correction is a simple description of the scattering by Neptune's hazy atmosphere. The residual scatter of 2.2 % in the cross-calibration is comparable to the contrast of many features.

Colour. The reference spectrum was measured four years after the encounter, and the filter passbands are approximated. The colour of the texture as it appears in the game is further affected by the game's lighting and tone curve.

Validation. The positions of features in the map have not been compared quantitatively with the positions tracked by Sromovsky et al. (1993); such a comparison would provide an independent test of the pointing and of the advection.

7 Conclusions

We have constructed a global map of Neptune's albedo and colour from 34 Voyager 2 narrow-angle camera images, with the pointing of each image determined from its limb and from the zonal symmetry of the cloud bands, a photometric cross-calibration through the overlaps of the images, and a colour derived from a ground-based spectrum. The map covers 85 % of the surface with data at 62 to 70 km per pixel. Its value lies less in the novelty of any step than in its documentation, its reproducibility from public data, and its open licence.

Data and code availability

The map, the scripts that produce it from the archived data, and the source of this note are available in the OpenReliant mods collection, under `sources/realistic-planets`: the list of

images with their geometry and download addresses, and Python scripts using NumPy, SciPy and ImageMagick for the download, the pointing, the fixed-pattern correction, the mosaics and the composition, which `build.sh` runs in order. The map and this note are released under the Creative Commons Attribution 4.0 licence, crediting OpenReliant and NASA/JPL; the scripts are released under the Mozilla Public License 2.0.

Acknowledgements

This work used data from the NASA Planetary Data System: the Voyager 2 images provided by the Ring-Moon Systems Node and the spectra of E. Karkoschka provided by the Atmospheres Node. The Voyager images are courtesy of NASA/JPL-Caltech. The processing pipeline and this note were developed with the assistance of an AI model (Claude, by Anthropic), under the author's direction.

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